

Problem Statement 63

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1. System Overview

The Community Solar Credit Allocation Manager is a clean-energy sharing portal that ingests rooftop solar generation metrics, facilitates peer-to-peer solar credit transfers within a neighborhood microgrid, and tracks monthly utility billing offsets.

Target stakeholders / actors:

- Prosumer Resident
- Co-op Manager

2. Requirements Table

ID	Priority	Requirement	Description	Acceptance Criteria	Rationale
FR-001	High	Solar Generation Recording	The system shall record solar generation units (kWh) from smart meter feeds and associate the generation data with the corresponding prosumer account.	Pass: Valid smart meter generation data is recorded against the correct household/account. Fail: Invalid or unassociated meter data is stored as valid generation.	This implements the source requirement to ingest rooftop solar generation metrics.
FR-002	High	Solar Credit Transfer	The system shall allow prosumers to transfer available surplus solar credits to neighbor accounts within the neighborhood microgrid.	Pass: Transferred credits are deducted from the prosumer balance and applied to the neighbor account. Fail: A transfer exceeding the available credit balance is accepted.	Peer-to-peer transfer of surplus credits is a core function explicitly identified in the problem statement.
FR-003	High	Credit Balance Management	The system shall maintain and display the current solar credit balance for each prosumer account after generation recording and credit transfers.	Pass: The displayed balance reflects all accepted generation and transfer transactions. Fail: The balance does not match the recorded transactions.	Accurate balances are required to prevent invalid credit transfers and support credit allocation.
FR-004	High	Monthly Billing Offset Tracking	The system shall calculate and track monthly utility billing offsets resulting from solar credits applied to each participating	Pass: The monthly statement shows the applicable solar-credit offset for the account. Fail: Applied credits are omitted or incorrectly	The source problem explicitly requires tracking monthly utility billing offsets.

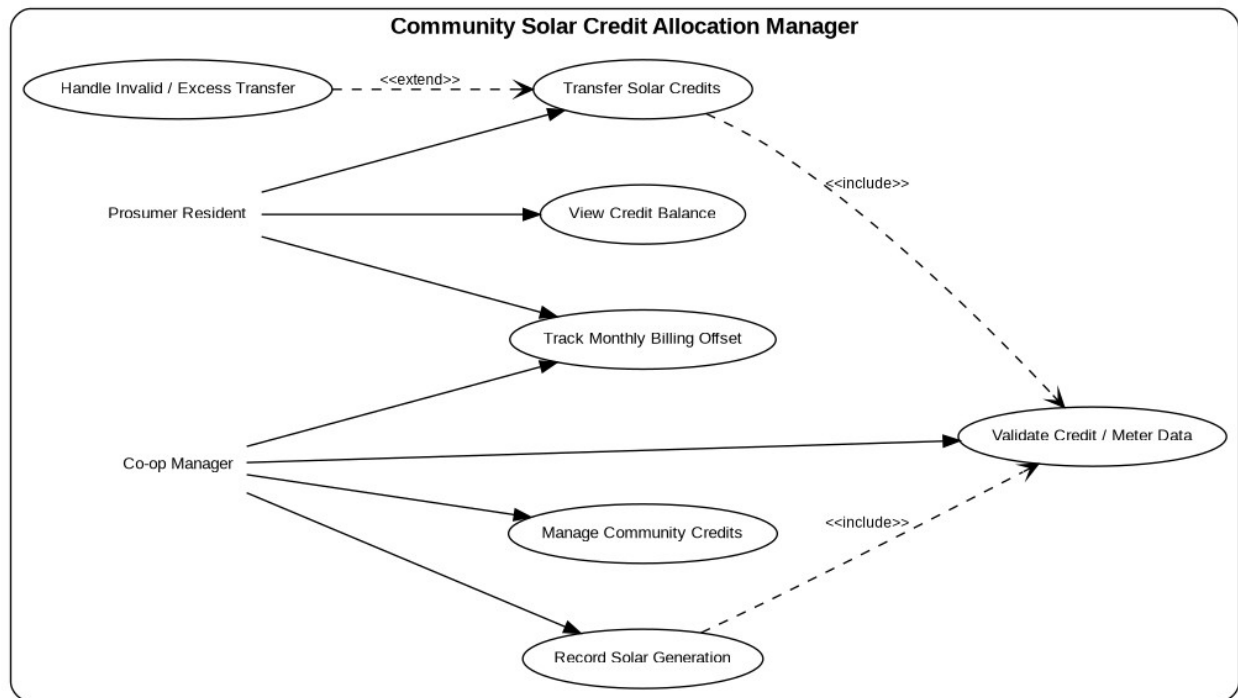
			account.	reflected in the monthly offset.	
FR-005	Medium	Community Credit Oversight	The system shall allow the Co-op Manager to review community solar generation, credit allocations, transfers, and monthly billing-offset information.	Pass: The Co-op Manager can retrieve current community-level records and transaction information. Fail: Required community credit records cannot be reviewed.	The Co-op Manager is a target actor and requires oversight of the neighborhood credit allocation process.

3. Non-Functional Requirements

ID	Type	Priority	Description	Acceptance Criteria	Rationale
NFR-001	Performance & Security	High	The smart meter telemetry processing pipeline shall validate and store hourly generation data for 500 households with 99.9% uptime.	Pass: Benchmarking tests confirm the required processing performance, validation behavior, and 99.9% availability under simulated peak load. Fail: The target household volume or availability requirement is not met.	This directly follows the supplied NFR guideline and ensures reliable ingestion of smart meter telemetry.
NFR-002	Security & Data Integrity	High	Solar credit balances, transfers, generation records, and billing-offset records shall be protected against unauthorized modification and inconsistent updates.	Pass: Security and integrity tests show unauthorized users cannot modify records and valid transactions preserve consistent balances and offsets. Fail: Unauthorized or inconsistent changes are accepted.	Credit allocation and billing offsets depend on trustworthy transaction and account data.

4. UML Use-Case Diagram

The diagram models both actors specified in the problem statement and the primary use cases. It contains «include» and «extend» relationships as required.



Relationship notes:

- Record Solar Generation «include» Validate Credit / Meter Data.
- Transfer Solar Credits «include» Validate Credit / Meter Data.
- Handle Invalid / Excess Transfer «extend» Transfer Solar Credits when a transfer fails validation, such as exceeding the available balance.

5. Use-Case Flow Specification — Transfer Solar Credits

Core use case: Transfer Solar Credits

Primary Actor	Prosumer Resident
Preconditions	1. The prosumer has an active account. 2. A current solar credit balance is available. 3. The recipient is a valid neighbor account within the neighborhood microgrid. 4. The transfer amount is positive and does not exceed the available transferable credit balance.
Postconditions	1. The requested credits are deducted from the sender's balance. 2. The transferred credits are applied to the recipient's account. 3. The transaction is recorded for community and billing-offset tracking. 4. Updated balances are available to the affected accounts.
Main Success Scenario	1. The prosumer selects a valid neighbor account as the recipient. 2. The prosumer enters the number of solar credits to transfer. 3. The system retrieves the prosumer's current credit balance. 4. The system validates the transfer amount and recipient account. 5. The system deducts the transferred credits from the prosumer balance. 6. The system applies the credits to the neighbor account. 7. The system records the transfer transaction. 8. The system updates the affected account balances and makes the transferred credits available for monthly billing-offset tracking.
Alternate Flow	A1. If the requested transfer exceeds the prosumer's available credit balance at Step 4, the system rejects the transfer. A2. No credits are deducted from the prosumer and no credits are applied to the recipient. A3. The system reports that the transfer cannot be completed because the available balance is insufficient. A4. The prosumer may enter a lower transfer amount and retry.

6. Submission Checklist

- Exactly 5 Functional Requirements: FR-001 to FR-005.
- Exactly 2 Non-Functional Requirements: NFR-001 and NFR-002.
- Requirements include the requested ID, Type/Priority, Description, Acceptance Criteria, and Rationale fields.
- UML use-case diagram models the Prosumer Resident and Co-op Manager and includes at least one «include» and one «extend» relationship.
- Use-case flow specification includes Preconditions, Postconditions, Main Success Scenario, and one Alternate Flow.

Source note: This solution is based on the uploaded Problem Statement #63. □filecite□turn1file0□